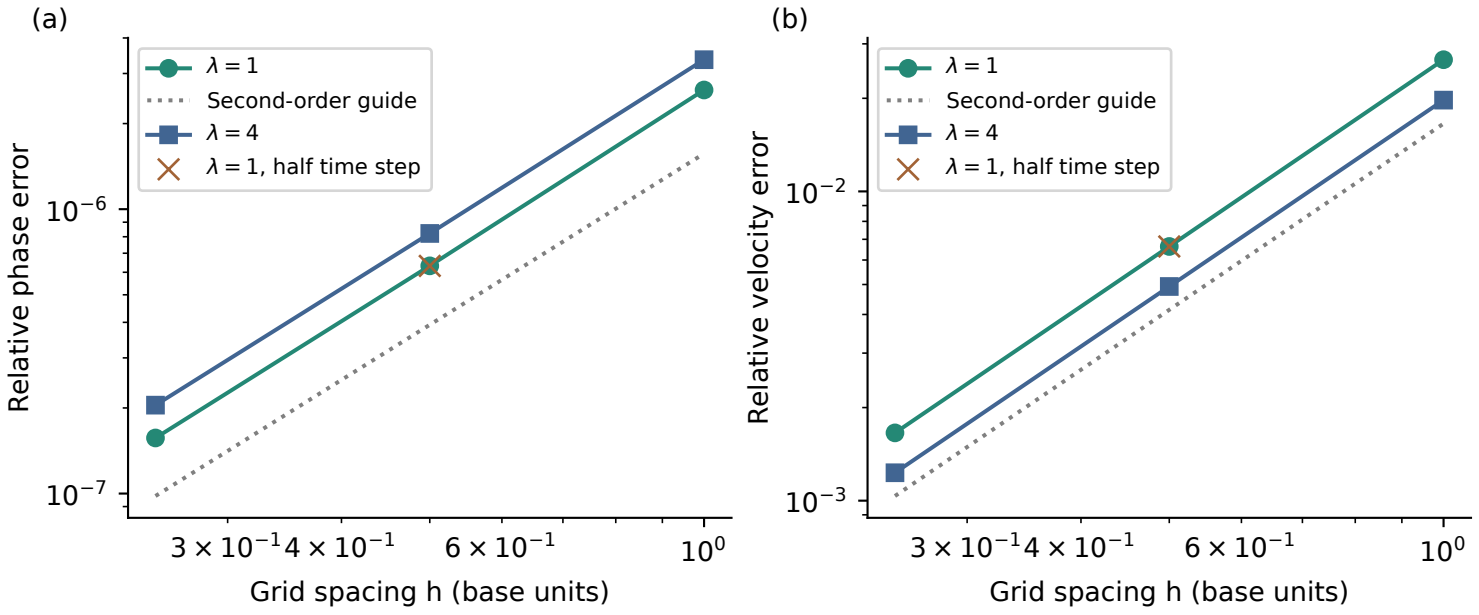
(a)
Relative phase error
10^{-6}
10^{-7}
$\lambda = 1$
Second-order guide
$\lambda = 4$
$\lambda = 1$, half time step
3×10^{-1}
4×10^{-1}
6×10^{-1}
10^{0}
Grid spacing h (base units)
(b)
Relative velocity error
10^{-2}
10^{-3}
$\lambda = 1$
Second-order guide
$\lambda = 4$
$\lambda = 1$, half time step
3×10^{-1}
4×10^{-1}
6×10^{-1}
10^{0}
Grid spacing h (base units)